Dry Ice Inventory Optimization Report

Executive Summary

This report provides a comprehensive analysis of the dry ice inventory for the period from 01-Jul-2024 to 30-Jun-2025. The analysis identifies significant opportunities for cost savings and operational efficiency. By implementing an EOQ-based inventory policy, an estimated KSh 16,751 can be saved monthly, representing a 81.0% reduction in transport costs.

Key Performance Indicators (KPIs)

Potential Monthly Savings	KSh 16,751
Economic Order Quantity (EOQ)	1,790 kg
Recommended Safety Stock	193 kg
Reorder Point	1,983 kg
Average Monthly Demand	4,043 kg

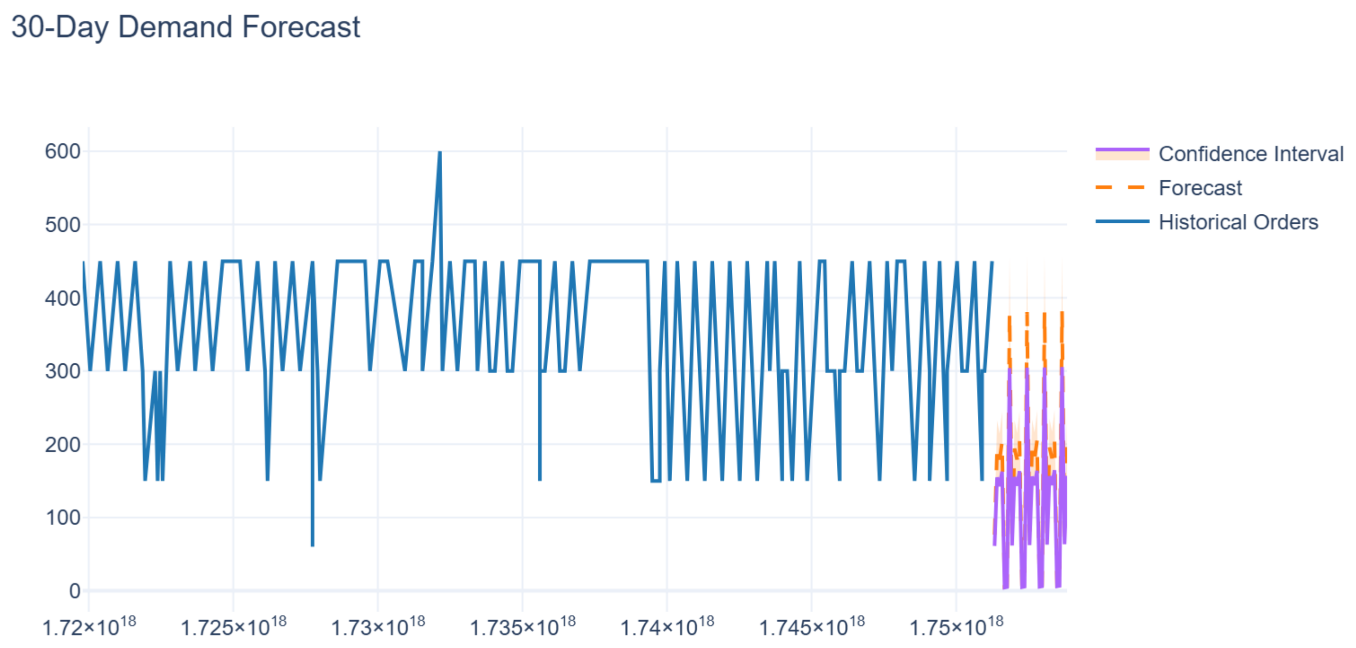
Top Recommendations

• Adopt an EOQ-based ordering policy. Place orders for **1790 kg** at a time to minimize total inventory costs.

• Establish a safety stock level of **193 kg**. This buffer will protect against demand variability and prevent stockouts.

• Set a reorder point at **1983 kg**. A new order should be triggered automatically when inventory falls to this level.

Demand Forecast Analysis



The 30-day demand forecast predicts fluctuating but consistent usage. The shaded area represents the 95% confidence interval, indicating the likely range of future demand. This forecast is crucial for planning orders and avoiding stockouts.

Cost Optimization Breakdown

The chart above clearly illustrates the cost benefits of switching to an EOQ-based system. While holding costs increase slightly due to larger average inventory, the reduction in ordering costs is substantial, leading to significant overall savings.